Lecture 2: Optimal Classifier

Spring 2026

10 ECTS Credits

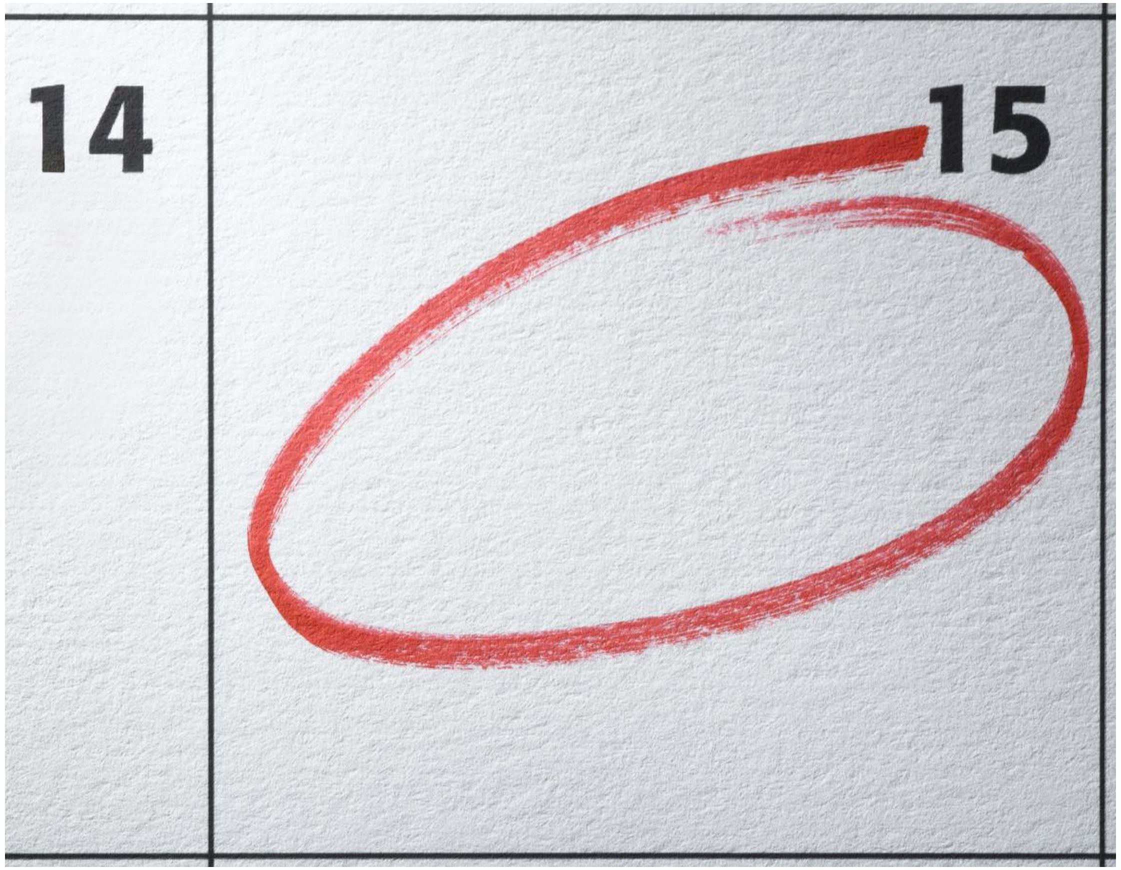
Department of Information and Communication Technology

Faculty of Engineering and Science

UiA

Recap

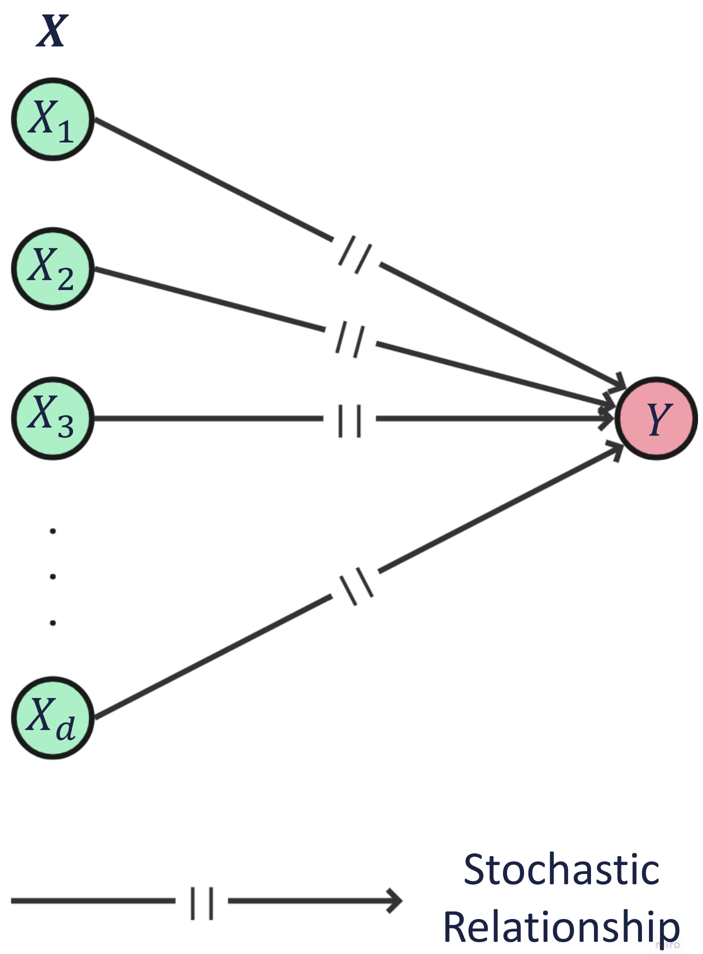
- Pattern recognition pipeline
- Single feature classification
- Decision threshold
- Two feature classification
- Some problems in finding an optimal model
  - Outliers and missing values
  - Feature extraction
  - Costs and risks
- Learning
  - Underfitting vs. Overfitting
  - Types of Learning



The problem

- Problem → feature vector and target
- $\boldsymbol{X}, Y$ relationship is stochastic in nature
- The relationship can only be specified by a joint feature-target distribution, $P_{\boldsymbol{X},Y}$.
- However:
  - Y is dependent on hidden/latent factors.
  - There are measurement noise in $\boldsymbol{X}$.

$\boldsymbol{X} \in R^d, Y \in R$



Prediction

- Supervised learning objective: **predict** Y given $\boldsymbol{X}$.
- Predictor, $\psi: \mathbb{R}^d \to \mathbb{R}$, such that $\psi(\boldsymbol{X})$ predicts Y and is designed using either:
  - Direct knowledge about $P_{\boldsymbol{X},Y}$, which can yield $\psi^*(\boldsymbol{X})$.
  - Indirect knowledge about $P_{\boldsymbol{X},Y}$ through an independent and identically distributed (i.i.d.) sample $S_n = \{(\boldsymbol{X}_1, Y_1), \ldots, (\boldsymbol{X}_n, Y_n)\}$
- A **loss function** $\ell: \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ defines the validity or optimality of a predictor.
- Loss functions like **quadratic loss**, **absolute difference loss**, **misclassification loss.**
- Since we deal with stochastic model, **prediction error** is given by:

$$L[\psi] = E\left[\ell\left(Y, \psi(\boldsymbol{X})\right)\right]$$

- The optimal predictor:

$$\psi^* = \arg\min_{\psi \in \mathcal{P}} L[\psi]$$

Recall: Expectation (Expected Value)

- Is expectation operation same as the average/mean?
- Expectation is the theoretical average of a random variable across its entire probability distribution:

$$E[X] = \begin{cases} \sum_x x \cdot P(X = x), & \text{(discrete)} \\ \int_{-\infty}^{\infty} x \cdot f(x) dx, & \text{(continuous)} \end{cases}$$

- $P(X = x)$ is the probability of $X = x$.
- $f(x)$ is the probability density function (PDF) of the continuous random variable.

Recall: Expectation (Expected Value)

- $X \rightarrow$ Outcome of rolling a fair die

$$E[X] = \sum_x x \cdot P(X = x)$$

$$E[X] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = 3.5$$

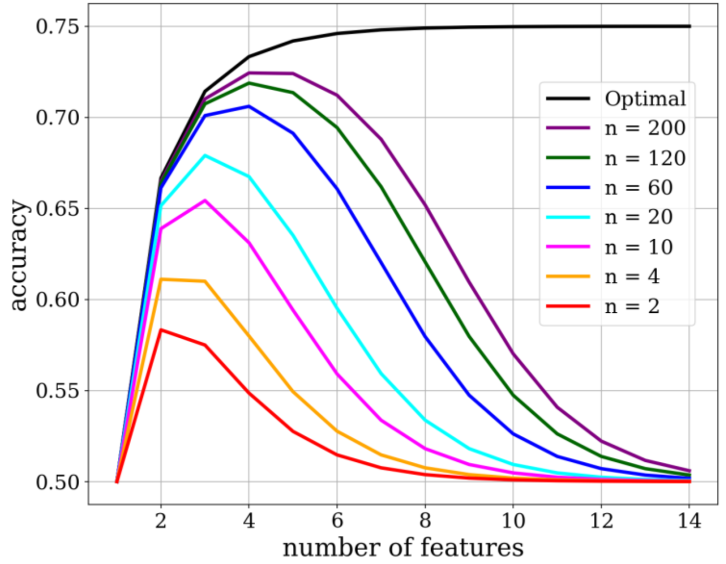
- If you roll $\{5, 2, 6, 2, 2, 1, 2, 3, 6, 1\}$, sample mean (average) over $N = 10$ rolls:

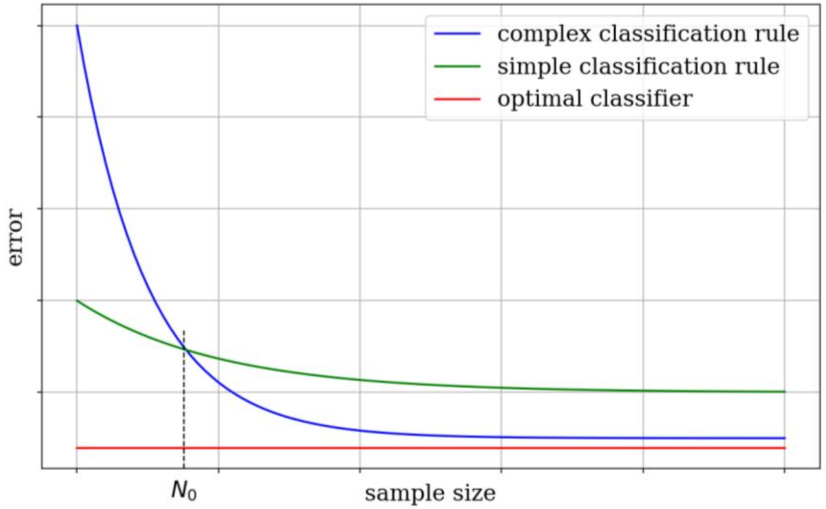
$$\bar{x} = (5 + 2 + 6 + 2 + 2 + 1 + 2 + 3 + 6 + 1)/10 = 3.0$$

- We can say the sample mean is 3.0 with a distance of 0.5 from the expected value of 3.5.
- However, as $N \rightarrow \infty$, $\bar{x} \rightarrow E[X]$.

Complexity Trade-offs

- **Curse of dimensionality** or Hughes Phenomenon:
  - Accuracy increases and then decreases as number of features increases.
  - The peaks shift right with increased sample size
- Seen in **scissors plot**:
  - Complex classifier: error converges to an optimal error with sample size.
  - Simple classifier: performs better under small sample sizes.
  - A problem dependent N_0, under which the simpler classifier performs better.





Classification without features

- Joint feature-label distribution $P_{\mathbf{X},Y}$ available → **optimal classifier** can be obtained.
- Simple case: a binary classifier with no features, $\hat{Y}$, such that $Y \in \{0,1\}$
- $\hat{Y}$ assign the label of the most common class.
- Class **prior probabilities/prevalences**: $P(Y = 0)$ and $P(Y = 1)$.
- Assigning a label closest to the mean.
- Rare disease example

$$\hat{Y} = \begin{cases} 1, P(Y = 1) > P(Y = 0), \\ 0, P(Y = 1) \leq P(Y = 0). \end{cases}$$

$$P(Y = 0) + P(Y = 1) = 1$$

$$\hat{Y} = 1, iff. P(Y = 1) > 0.5$$

$$E[Y] = P(Y = 1)$$

$$\varepsilon[\hat{Y}] = P(\hat{Y} \neq Y) = \min\{P(Y = 1), P(Y = 0)\}$$

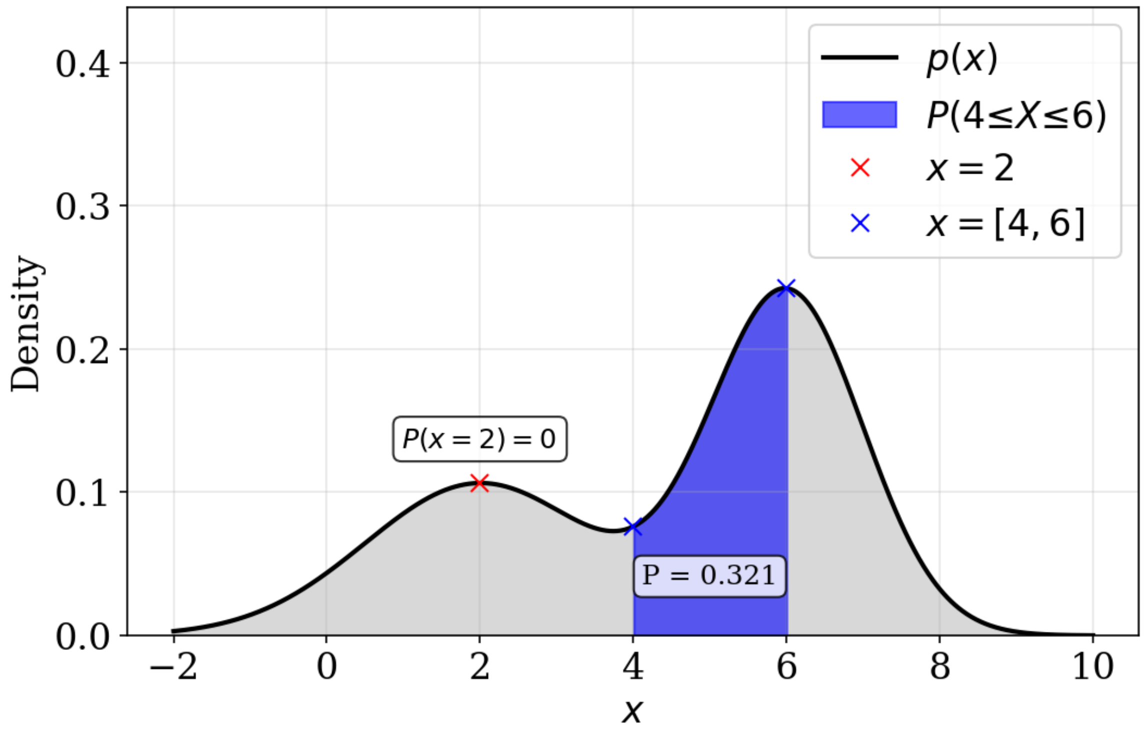
Recall: Probability Density Functions (PDFs)

- Let us consider a task with a single continuous feature $x \in \mathbb{R}$ and plot its PDF, $p(x)$.
  - What does this plot show?
  - What is $\mathrm{P}(x = 2)$?

$$P(x = 2) = 0$$

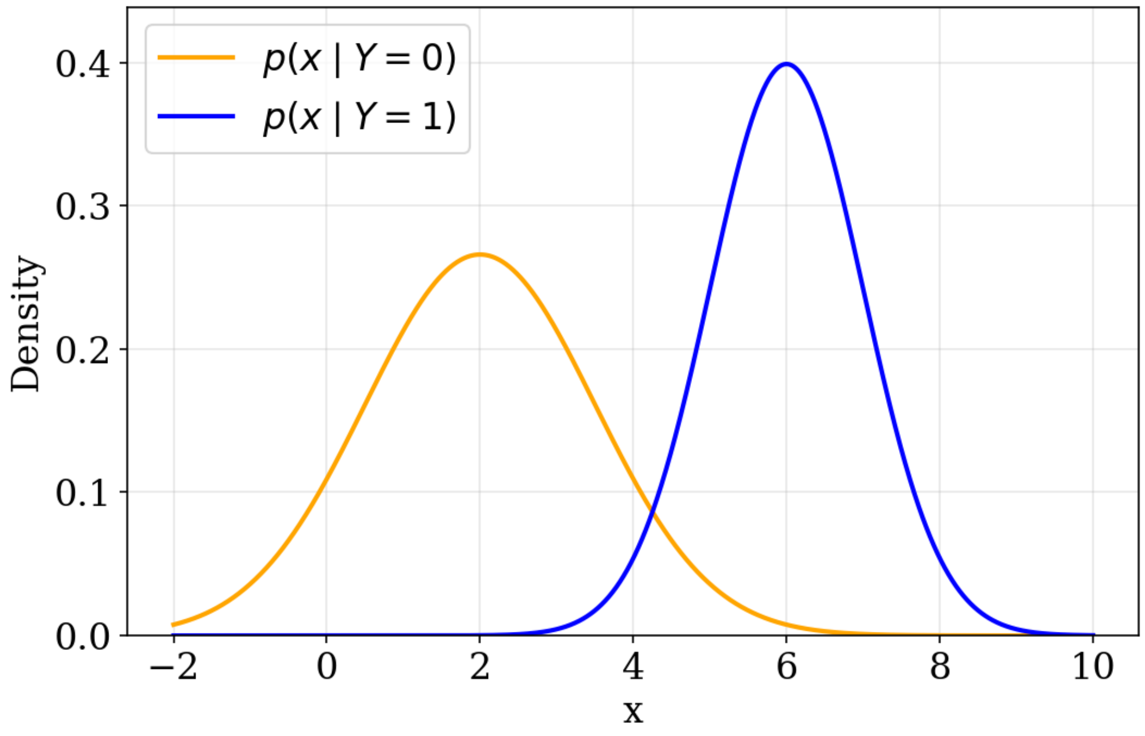
  - What is $\mathrm{P}(4 \leq x \leq 6)$?

$$\mathrm{P}(4 \leq x \leq 6) = \int_E p(x)dx = 0.321$$



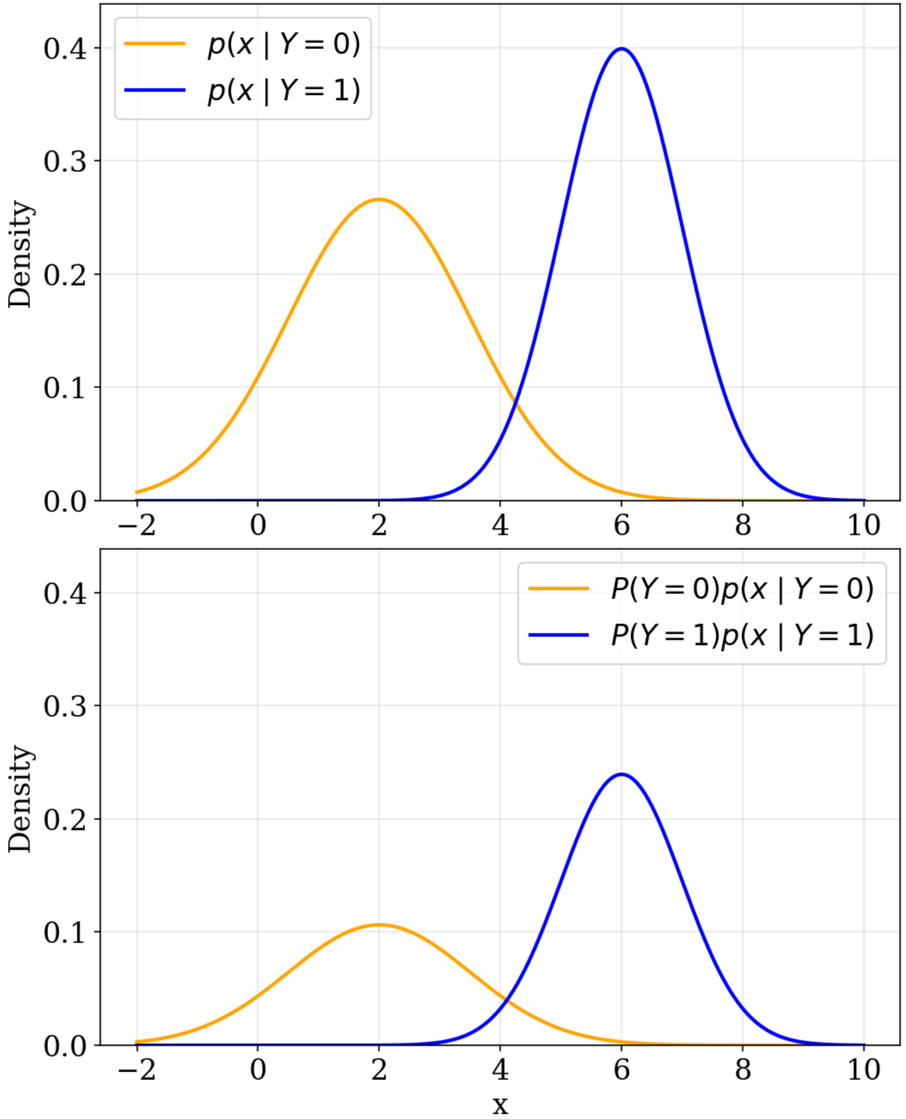
Recall: Probability Density Functions (PDFs)

- What are **class-conditional densities**?
- If I already know this instance belongs to class 0 (or 1), what feature values am I likely to see?
  - $p(x \mid Y = 0) \rightarrow$ Distribution of the feature among class 0 instances.
  - $p(x \mid Y = 1) \rightarrow$ Distribution of the feature among class 1 instances.
- However, class-conditional densities doesn't account for how often each class occurs.



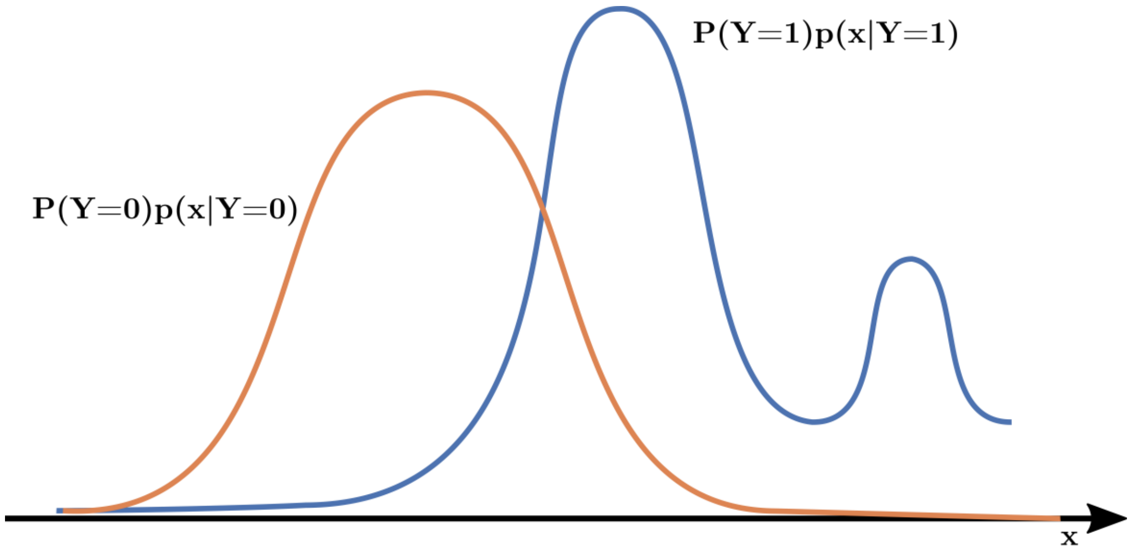
Recall: Probability Density Functions (PDFs)

- The class-conditional densities are weighted by the priors.
$$p(x, Y = 0) = P(Y = 0)p(\, x \mid Y = 0 \,)$$
$$p(x, Y = 1) = P(Y = 1)p(\, x \mid Y = 1 \,)$$
- These curves represent the actual contribution of classes to the overall feature distribution.
- If we add these together, we get:
$P(Y = 0)p(\, x \mid Y = 0 \,) + P(Y = 1)p(\, x \mid Y = 1 \,)$
$= p(x)$
- This is known as the **marginal PDF** of the feature.
$$p(x) = \sum_{Y \in \{0,1\}} p(x, y)$$



Classification with features

- Each $\boldsymbol{X} \in \mathbb{R}^d$ is a **feature vector**,
  - where d, is the **dimension**,
  - and $\mathbb{R}^d$ is called the **feature space**.
- We assume all features are real-valued and continuous.
- **Class conditional densities** $p(\boldsymbol{x}|Y=0)$ and $p(\boldsymbol{x}|Y=1)$.
- When $E = R^d$, LHS becomes prior probabilities.
- Marginal PDF $p(\boldsymbol{x})$:
- $P_{\boldsymbol{X},Y}$ is fully specified by the class-conditional densities weighted by prior probabilities.



$$\boldsymbol{X} = [X_1, X_2, \dots, X_d] \in \mathbb{R}^d$$

$$\mathcal{C} = \{0,1\}$$

$$P(\boldsymbol{X} \in E, Y = c) = P(Y = c) \int_E p(\boldsymbol{x}|Y = c) d\boldsymbol{x}$$

$$E \subseteq \mathbb{R}^d$$

$$p(\boldsymbol{x}) = P(Y = 0) p(\boldsymbol{x}|Y = 0) + P(Y = 1) p(\boldsymbol{x}|Y = 1)$$

Cookie Quality Control

- In a bakery, 5% cookies are perfectly baked (95% are not).
- A human inspector flags
  - 85% of perfectly baked cookies correctly,
  - 12% of imperfect cookies wrong.
- The question:

If the inspector says a cookie is perfect, what's the actual chance it really is perfect?

- Answer: approx. **27%**. Why?

Cookie Quality Control

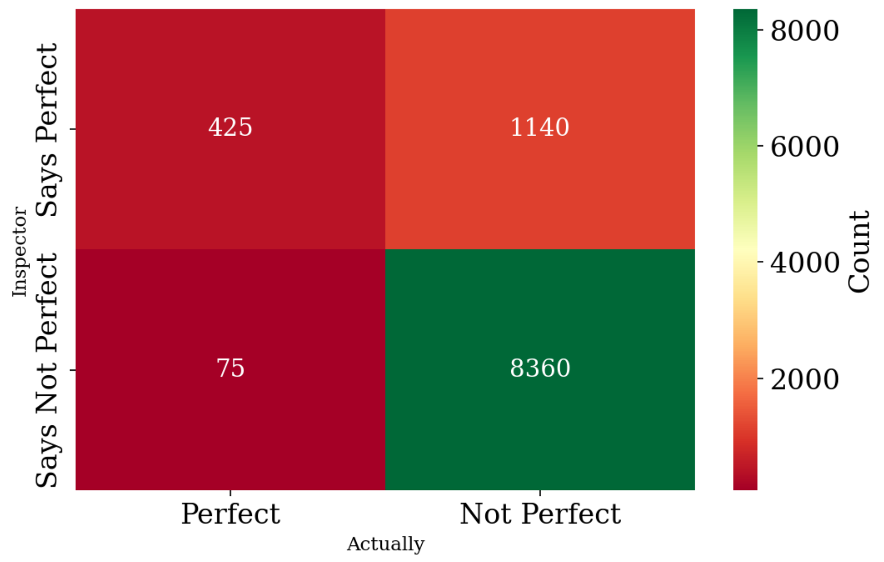
	Actually Perfect (5%)	Not Perfect (95%)
Inspector Says "Perfect"	85% (True Positive)	12% (False Positive)
Inspector Says "Not Perfect"	15% (False Negative)	88% (True Negative)

If the inspector says a cookie is perfect, what's the actual chance it really is perfect?

- Let us say, 10000 cookies are tested a month:

$$P(\text{Perfect} \mid \text{Says Perfect}) = \frac{425}{425 + 1140} \approx 27.16\%$$

$$P(\text{Perfect} \mid \text{Says Perfect}) = \frac{P(\text{Says Perfect} \mid \text{Perfect}) \times P(\text{Perfect})}{P(\text{Says Perfect})}$$



- Why the very small value?

Bayes' Theorem

- The theorem states that:

$$P(\mathrm{H} \mid \mathrm{E}) = \frac{P(\mathrm{E} \mid \mathrm{H})P(H)}{P(E)} = \frac{P(\mathrm{E} \mid \mathrm{H})P(H)}{P(\mathrm{E} \mid \mathrm{H})\mathrm{P}(\mathrm{H}) + P(\mathrm{E} \mid \text{not H})P(\text{not H})}$$

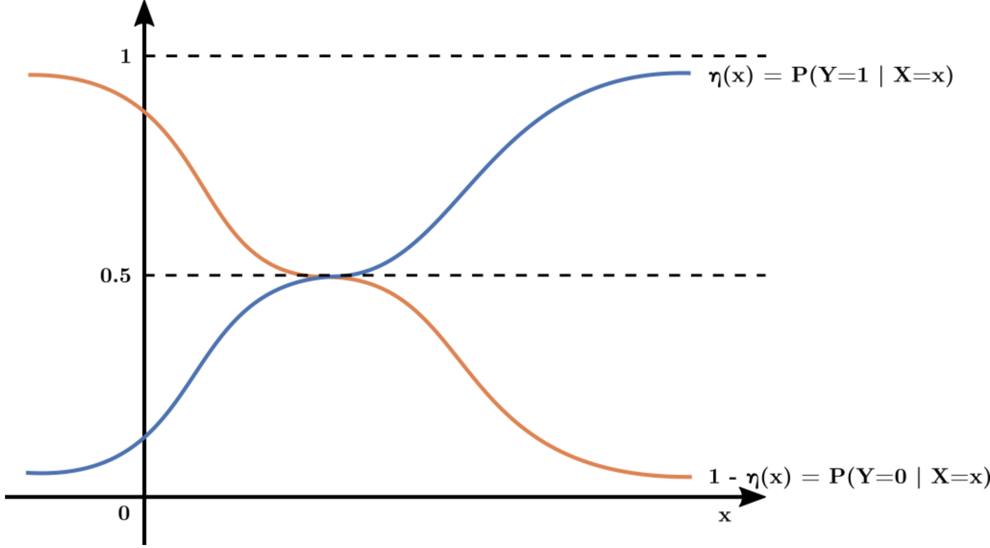
$$\mathrm{H} \rightarrow \text{Hypothesis}, \qquad \mathrm{E} \rightarrow \text{Event}$$

- Or more generally for two events A and B and $P(B) \neq 0$:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

  - $P(A \mid B) \rightarrow$ Conditional probability of A occurring given B is true.
  - $P(B \mid A) \rightarrow$ Conditional probability of B occurring given A is true.
  - $P(A)$ and $P(B)$ are probabilities of A and B occurring independently.

Posterior probabilities



- Assuming $p(\boldsymbol{x}) > 0$:
- **Posterior probability** of Y given $\boldsymbol{X} = \boldsymbol{x}$:

$$P(Y=0|\boldsymbol{X}=\boldsymbol{x}) = \frac{P(Y=0)p(\boldsymbol{x}|Y=0)}{p(\boldsymbol{x})} = \frac{P(Y=0)p(\boldsymbol{x}|Y=0)}{P(Y=0)p(\boldsymbol{x}|Y=0)+P(Y=1)p(\boldsymbol{x}|Y=1)}$$

$$P(Y=1|\boldsymbol{X}=\boldsymbol{x}) = \frac{P(Y=1)p(\boldsymbol{x}|Y=1)}{p(\boldsymbol{x})} = \frac{P(Y=1)p(\boldsymbol{x}|Y=1)}{P(Y=0)p(\boldsymbol{x}|Y=0)+P(Y=1)p(\boldsymbol{x}|Y=1)}$$

  - Value ranges between 0 and 1, and
  - Sum up to 1, for the entire feature space.
- Thus, for a binary classification, we can use the **posterior probability function:**

$$\eta(\boldsymbol{x}) = E[Y|X=x] = P(Y=1|\boldsymbol{X}=\boldsymbol{x}), \boldsymbol{x} \in \mathbb{R}^d$$

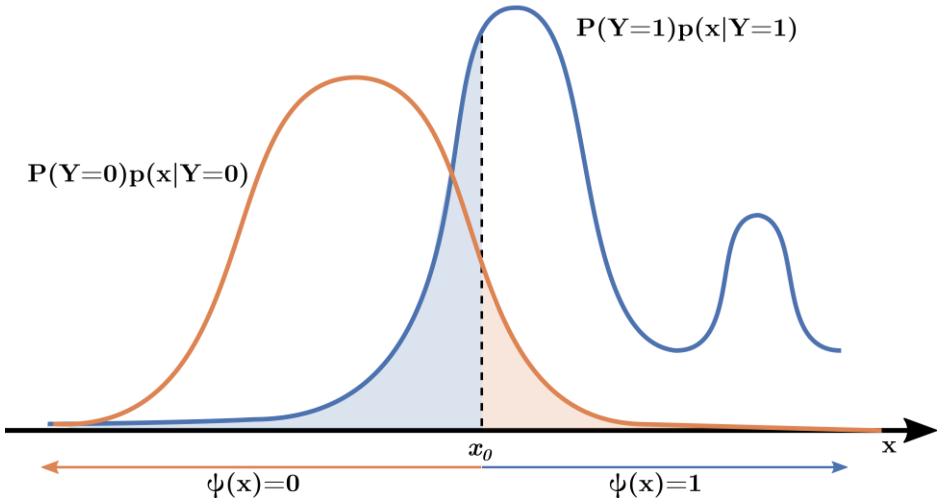
Classifiers and Classification Error

- A **classifier** is a function $\psi: \mathbb{R}^d \rightarrow \{0,1\}$ that divides the feature space R^d into two sets:
  - a 0-decision region, and
  - a 1-decision region

  by a **decision boundary**, x_o.
- Classification error is the probability of misclassification:

$$\varepsilon[\psi] = P(\psi(\boldsymbol{X}) \neq Y)$$

- Main criterion for performance in classification, determined by $P_{\boldsymbol{X},Y}$



Class-specific Errors

- **Class specific error rates** of ψ measures misclassification rates within each class:
  - **False Positive Rate**

$$\varepsilon^0[\psi] = P(\psi(\boldsymbol{X}) = 1|Y = 0) = \int_{\{x|\psi(x)=1\}} p(\boldsymbol{x}|Y = 0)d\boldsymbol{x}$$

  - **False Negative Rate**

$$\varepsilon^1[\psi] = P(\psi(\boldsymbol{X}) = 0|Y = 1) = \int_{\{x|\psi(x)=0\}} p(\boldsymbol{x}|Y = 1)d\boldsymbol{x}$$

- **Total error** (1):

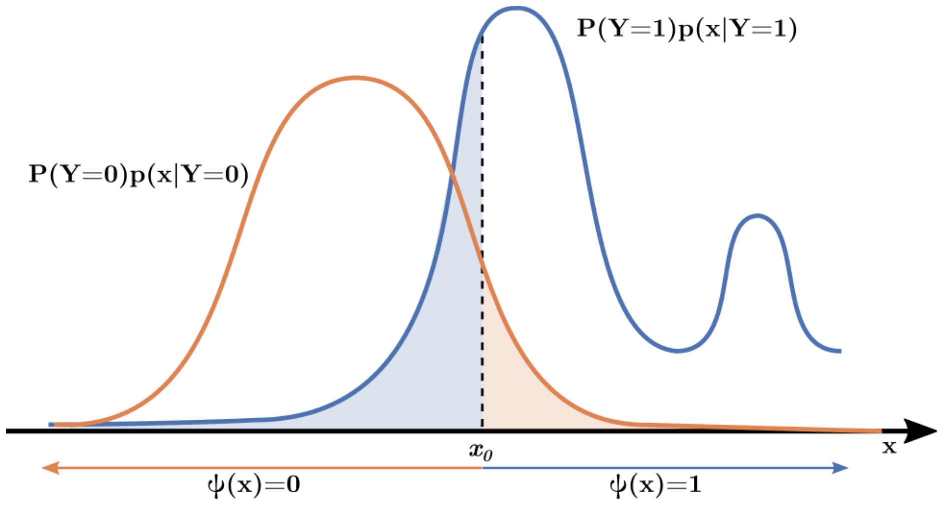
$$\varepsilon[\psi] = P(\psi(\boldsymbol{X}) \neq Y) = P(\psi(\boldsymbol{X}) = 1, Y = 0) + P(\psi(\boldsymbol{X}) = 0, Y = 1)$$

$$= P(Y = 0)P(\psi(\boldsymbol{X}) = 1|Y = 0) + P(Y = 1)P(\psi(\boldsymbol{X}) = 0|Y = 1)$$

$$= P(Y = 0)\int_{\{x|\psi(x)=1\}} p(\boldsymbol{x}|Y = 0)d\boldsymbol{x} + P(Y = 1)\int_{\{x|\psi(x)=0\}} p(\boldsymbol{x}|Y = 1)d\boldsymbol{x}$$

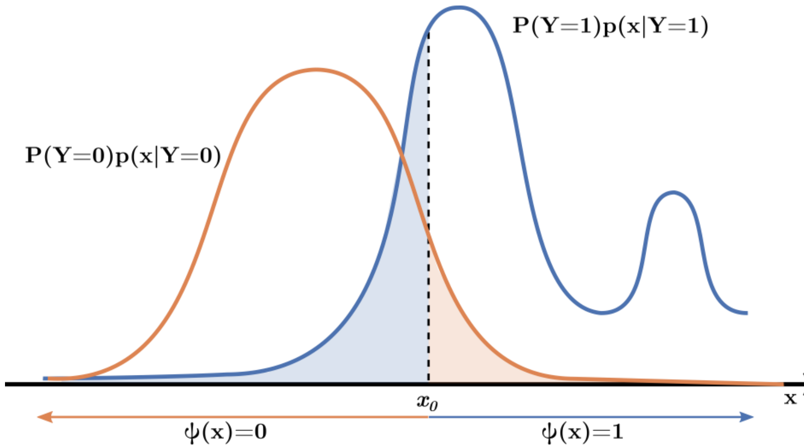
$$= (1 - n)\varepsilon^0[\psi] + n\varepsilon^1[\psi]$$

- A linear combination of class-specific error rates weighted by class priors.



Conditional Classification Error

- **Conditional error** of a classifier:
$$\varepsilon[\psi|\boldsymbol{X}=\boldsymbol{x}] = P(\psi(\boldsymbol{X}) \neq Y|\boldsymbol{X}=\boldsymbol{x})$$
- Using the **Law of Total Probability for random variables**, classification error is the expected conditional classification error over the feature space (2):
$$\varepsilon[\psi] = E[\varepsilon(\psi|\boldsymbol{X}=\boldsymbol{x})] = \int_{\boldsymbol{x}\in\mathbb{R}^d} \varepsilon[\psi \mid \boldsymbol{X}=\boldsymbol{x}]p(\boldsymbol{x})d\boldsymbol{x}$$
- The conditional classification error $\varepsilon[\psi \mid \boldsymbol{X}=\boldsymbol{x}]$ is determined by the posterior probability function $\eta(\boldsymbol{x})$ as (3):
$$\begin{aligned}\varepsilon[\psi \mid \boldsymbol{X}=\boldsymbol{x}] &= P(\psi(\boldsymbol{X})=0, Y=1) \mid \boldsymbol{X}=\boldsymbol{x}) + P(\psi(\boldsymbol{X})=1, Y=0) \mid \boldsymbol{X}=\boldsymbol{x}) \\ &= I_{\psi(\boldsymbol{x})=0}P(Y=1 \mid \boldsymbol{X}=\boldsymbol{x}) + I_{\psi(\boldsymbol{x})=1}P(Y=0 \mid \boldsymbol{X}=\boldsymbol{x}) \\ &= I_{\psi(\boldsymbol{x})=0}\eta(\boldsymbol{x}) + I_{\psi(\boldsymbol{x})=1}(1-\eta(\boldsymbol{x})) = \begin{cases} \eta(\boldsymbol{x}), & \text{if } \psi(\boldsymbol{x})=0 \\ 1-\eta(\boldsymbol{x}), & \text{if } \psi(\boldsymbol{x})=1 \end{cases}\end{aligned}$$



- Recall, for the above derivation:
  - $\eta(\boldsymbol{x}) = P(Y=1 \mid \boldsymbol{X}=\boldsymbol{x})$: Probability that the true label is 1, given $\boldsymbol{X}=\boldsymbol{x}$.
  - $1-\eta(\boldsymbol{x}) = P(Y=0 \mid \boldsymbol{X}=\boldsymbol{x})$: Probability that the true label is 0, given $\boldsymbol{X}=\boldsymbol{x}$.

Bayes Classifier

- A **Bayes classifier** minimizes the classification error:

$$\psi^* = \arg\min_{\psi \in \mathcal{C}} P(\psi(\boldsymbol{X}) \neq Y)$$

- There maybe an infinite number of Bayes classifier for a given problem.
- **Theorem** (4): The classifier

$$\psi^*(\boldsymbol{x}) = \arg\max_i P(\, Y = i \mid \boldsymbol{X} = \boldsymbol{x}\,) = \begin{cases} 1, & \eta(x) > 0.5 \\ 0, & \text{otherwise} \end{cases}$$

  for $\boldsymbol{x} \in \mathbb{R}^d$ is a Bayes Classifier.
- Only the knowledge of the posterior probability function is necessary to obtain a Bayes Classifier.

Theorem: Proof

- For all $\psi \in \mathcal{C}$, we want to show that
$$\varepsilon[\psi] \geq \varepsilon^*[\psi]$$
- From (2)
$$\varepsilon[\,\psi \mid \boldsymbol{X} = \boldsymbol{x}\,] \geq \varepsilon[\,\psi^* \mid \boldsymbol{X} = \boldsymbol{x}\,], \forall\, \boldsymbol{x} \in \mathbb{R}^d$$
- Or,
$$\varepsilon[\,\psi \mid \boldsymbol{X} = \boldsymbol{x}\,] - \varepsilon[\,\psi^* \mid \boldsymbol{X} = \boldsymbol{x}\,] \geq 0, \forall\, \boldsymbol{x} \in \mathbb{R}^d$$
- Using (3),
$$\begin{aligned}
&\varepsilon[\,\psi \mid \boldsymbol{X} = \boldsymbol{x}\,] - \varepsilon[\,\psi^* \mid \boldsymbol{X} = \boldsymbol{x}\,] \\
&= I_{\psi(\boldsymbol{x})=0}\eta(\boldsymbol{x}) + I_{\psi(\boldsymbol{x})=1}\big(1-\eta(\boldsymbol{x})\big) - \Big(I_{\psi^*(\boldsymbol{x})=0}\eta(\boldsymbol{x}) + I_{\psi^*(\boldsymbol{x})=1}\big(1-\eta(\boldsymbol{x})\big)\Big) \\
&= \eta(\boldsymbol{x})\big(I_{\psi(\boldsymbol{x})=0} - I_{\psi^*(\boldsymbol{x})=0}\big) + \big(1-\eta(\boldsymbol{x})\big)\big(I_{\psi(\boldsymbol{x})=1} - I_{\psi^*(\boldsymbol{x})=1}\big)
\end{aligned}$$

Theorem: Proof

- Let us look at the possibilities in making up these differences:

$$I_{\psi(\boldsymbol{x})=0} - I_{\psi^*(\boldsymbol{x})=0}, \qquad I_{\psi(\boldsymbol{x})=1} - I_{\psi^*(\boldsymbol{x})=1}$$

- Exhausting all the possibilities for $\psi(\boldsymbol{x})$ and $\psi^*(\boldsymbol{x})$:

$$I_{\psi(\boldsymbol{x})=0} - I_{\psi^*(\boldsymbol{x})=0} = -\left(I_{\psi(\boldsymbol{x})=1} - I_{\psi^*(\boldsymbol{x})=1}\right)$$

$\psi(\boldsymbol{x})$	$\psi^*(\boldsymbol{x})$	$I_{\psi(\boldsymbol{x})=0}$	$I_{\psi^*(\boldsymbol{x})=0}$	$I_{\psi(\boldsymbol{x})=0} - I_{\psi^*(\boldsymbol{x})=0}$	$I_{\psi(\boldsymbol{x})=1}$	$I_{\psi^*(\boldsymbol{x})=1}$	$I_{\psi(\boldsymbol{x})=1} - I_{\psi^*(\boldsymbol{x})=1}$
0	0	1	1	0	0	0	0
0	1	1	0	+1	0	1	-1
1	0	0	1	-1	1	0	+1
1	1	0	0	0	1	1	0

Theorem: Proof

- Substituting $I_{\psi(\boldsymbol{x})=0} - I_{\psi^*(\boldsymbol{x})=0} = -\left(I_{\psi(\boldsymbol{x})=1} - I_{\psi^*(\boldsymbol{x})=1}\right)$,

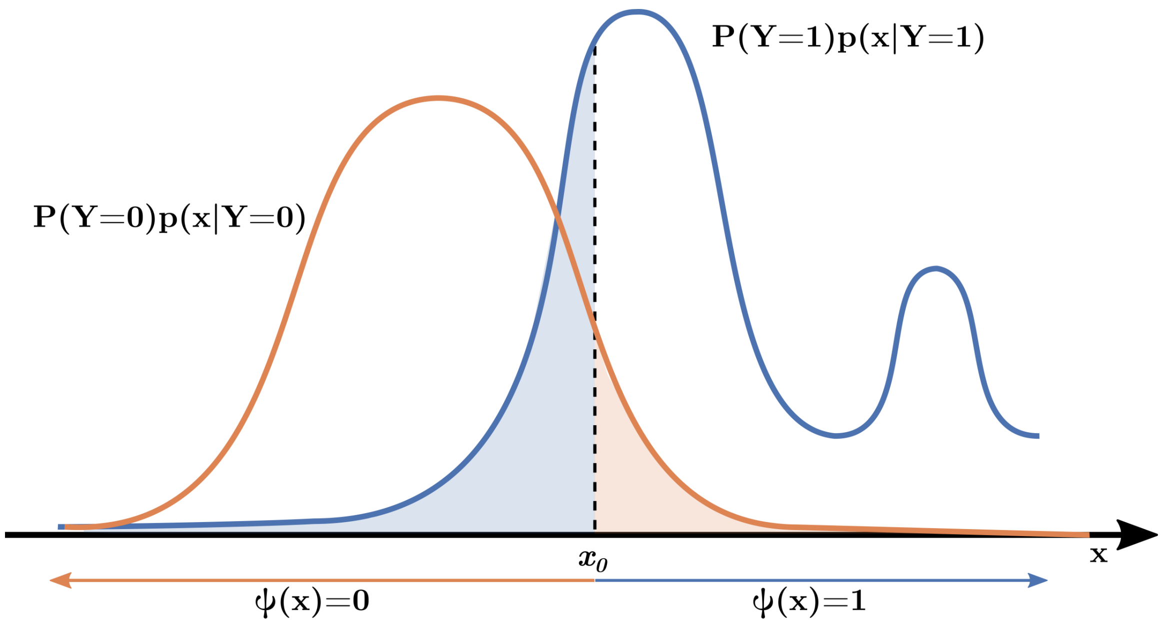
$$\begin{aligned} &\varepsilon[\,\psi \mid \boldsymbol{X} = \boldsymbol{x}\,] - \varepsilon[\,\psi^* \mid \boldsymbol{X} = \boldsymbol{x}\,] \\ &= \eta(\boldsymbol{x})\left(I_{\psi(\boldsymbol{x})=0} - I_{\psi^*(\boldsymbol{x})=0}\right) - \left(1 - \eta(\boldsymbol{x})\right)\left(I_{\psi(\boldsymbol{x})=0} - I_{\psi^*(\boldsymbol{x})=0}\right) \\ &= (2\eta(\boldsymbol{x}) - 1)\left(I_{\psi(\boldsymbol{x})=0} - I_{\psi^*(\boldsymbol{x})=0}\right) \end{aligned}$$

- There are only two possibilities:
  - $\eta(\boldsymbol{x}) > \frac{1}{2} \rightarrow$ Both terms in the parentheses are nonnegative.
  - $\eta(\boldsymbol{x}) \leq \frac{1}{2} \rightarrow$ Both terms in the parentheses are nonpositive.
- Thus, in both cases the product is nonnegative meaning:

$$\varepsilon[\,\psi \mid \boldsymbol{X} = \boldsymbol{x}\,] - \varepsilon[\,\psi^* \mid \boldsymbol{X} = \boldsymbol{x}\,] \geq 0, \forall\, \boldsymbol{x} \in \mathbb{R}^d$$

Bayes Classifier (Contd.)

- Optimal decision boundary is the set:
$$\{\boldsymbol{x} \in R^d \mid \eta(\boldsymbol{x}) = 0.5\}$$
- Bayes classifier is defined pointwise to minimize the error $\varepsilon[\,\psi \mid \boldsymbol{X} = \boldsymbol{x}\,]$ at each value of $\boldsymbol{x} \in R^d$, which minimizes the error globally.
- Bayes classifier is both local and global optimal predictor.



Bayes classifier (Contd.)

- Bayes classifier theorem for $x \in R^d$ can be restated as:

$$\psi^*(\boldsymbol{x}) = \begin{cases} 1, & \eta(\boldsymbol{x}) > 1 - \eta(\boldsymbol{x}) \\ 0, & \text{otherwise} \end{cases}$$

- Considering posterior probability equation:

$$\psi^*(\boldsymbol{x}) = \begin{cases} 1, & P(Y=1)p(\boldsymbol{x} \mid Y=1) > P(Y=0)p(\boldsymbol{x} \mid Y=0) \\ 0, & \text{otherwise} \end{cases}$$

- Bayes classifier can be determined by comparing at each point in feature space the weighted class-conditional densities.
- The above equation can be rewritten as:

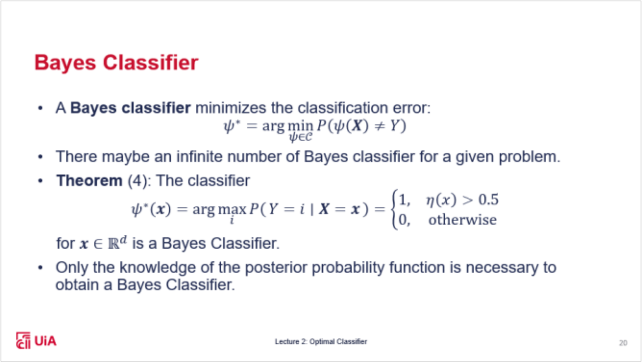
$$\psi^*(\boldsymbol{x}) = \begin{cases} 1, & D^*(\boldsymbol{x}) > k^* \\ 0, & \text{otherwise} \end{cases}$$

where the **optimal discriminant** $D^*: R^d \rightarrow R$ is given by

$$D^*(\boldsymbol{x}) = \ln \frac{p(\boldsymbol{x} \mid Y=1)}{p(\boldsymbol{x} \mid Y=0)}$$

for $x \in R^d$, with **optimal threshold** k^* given by

$$k^* = \ln \frac{P(Y=0)}{P(Y=1)}$$

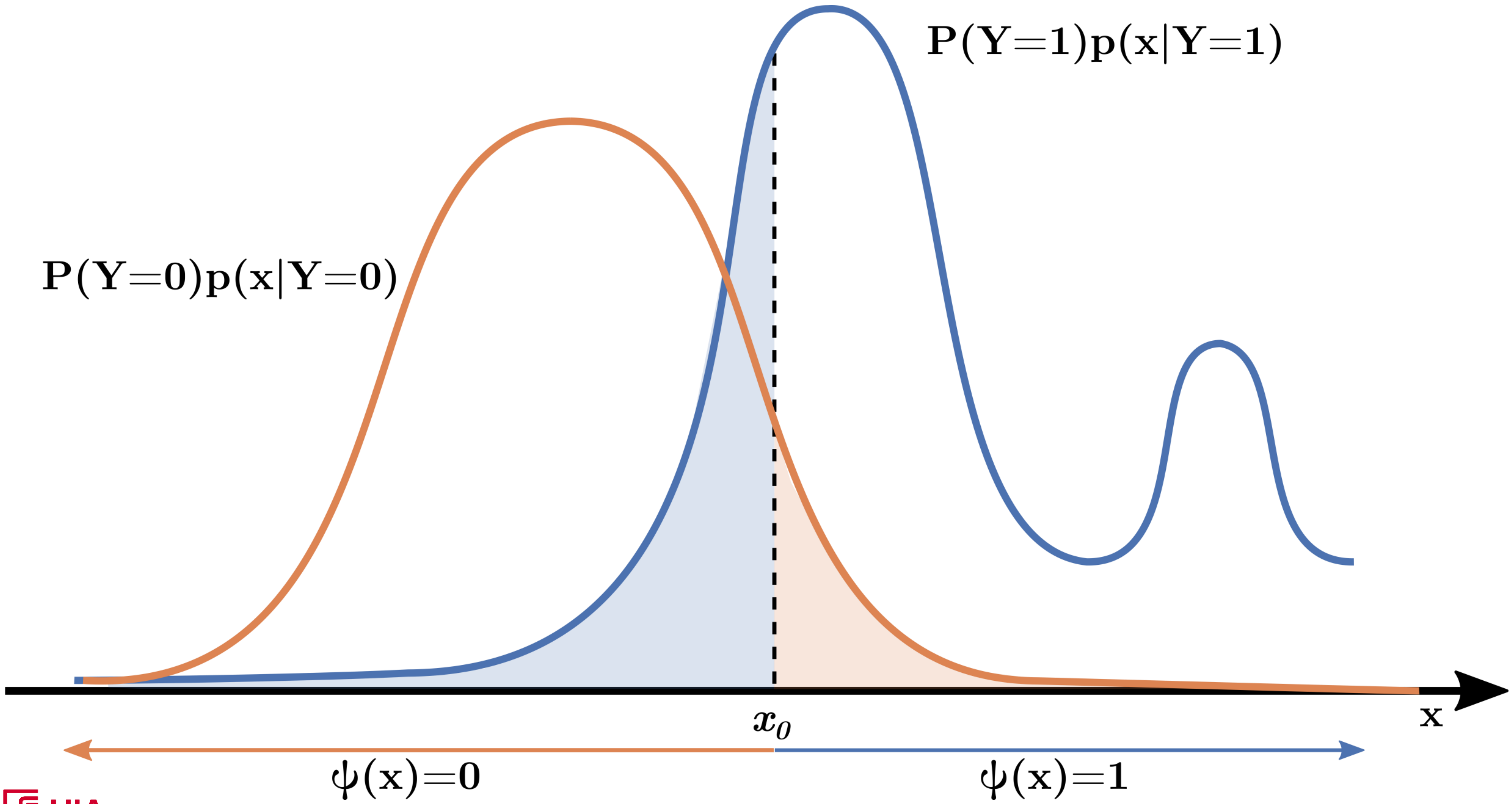


Bayes classifier (Contd.)

- Bayes Classifier:

$$\psi^*(\boldsymbol{x}) = \begin{cases} 1, & D^*(\boldsymbol{x}) > k^* \\ 0, & \text{otherwise} \end{cases}, \qquad D^*(\boldsymbol{x}) = \ln \frac{p(\boldsymbol{x} \mid Y = 1)}{p(\boldsymbol{x} \mid Y = 0)}, \qquad k^* = \ln \frac{P(Y = 0)}{P(Y = 1)}$$

- $D^*(\boldsymbol{x})$ is called the **log-likelihood ratio** or **optimal discriminant function**:
  - Measures how much more likely $\boldsymbol{x}$ is under class 1 vs. class 0.
  - With $D^*(\boldsymbol{x}) > k^*$, $\boldsymbol{x}$ provides enough evidence to predict class 1.
- k^* adjusts for **class imbalance**:
  - If Class 0 is more common $\big(P(Y = 0) > P(Y = 1)\big)$, $k^* > 0$. You need a stronger evidence $\big(D^*(\boldsymbol{x})\big)$ to predict class 1.
  - If $P(Y = 0) = P(Y = 1)$, $k^* = 0$, we compare likelihoods directly.


$\mathbf{P(Y{=}1)p(x|Y{=}1)}$
$\mathbf{P(Y{=}0)p(x|Y{=}0)}$
x_0
x
$\psi(\mathbf{x}){=}\mathbf{0}$
$\psi(\mathbf{x}){=}\mathbf{1}$

Bayes Error

- Given a Bayes classifier ψ^*, the error $\varepsilon^* = \varepsilon[\psi^*]$ is known as **Bayes error.**
- All Bayes classifiers share the same Bayes error: the lower bound on the classification error that can be achieved in a given problem.
- Should be sufficiently small to design a good classifier from data.
- **Theorem**: In two-class classification problem, the Bayes error is given by:

$$\varepsilon^* = E[\min\{\eta(\boldsymbol{X}), 1 - \eta(\boldsymbol{X})\}]$$

  Further, the maximum value Bayes error can take is 0.5.

Bayes Error – Proof

- From (2), (3), and (4):

$$\varepsilon^* = E[\varepsilon(\psi^*|\boldsymbol{X}=\boldsymbol{x})] = E\left[I_{\psi^*(\boldsymbol{x})=0}\,\eta(\boldsymbol{x}) + I_{\psi^*(\boldsymbol{x})=1}\big(1-\eta(\boldsymbol{x})\big)\right]$$

$$= E\left[\mathrm{I}_{\eta(\boldsymbol{x})\le 1-\eta(\boldsymbol{x})}\,\eta(\boldsymbol{X}) + \mathrm{I}_{\eta(\boldsymbol{x})>1-\eta(\boldsymbol{x})}\big(1-\eta(\boldsymbol{X})\big)\right] = E[\min\{\eta(\boldsymbol{X}), 1-\eta(\boldsymbol{X})\}]$$

- Applying the identity $\min\{a, 1-a\} = \frac{1}{2} - \frac{1}{2}|2a-1|, \forall\, 0 \le a \le 1$

$$\varepsilon^* = \frac{1}{2} - \frac{1}{2}E[|2\eta(\boldsymbol{X})-1|]$$

$$\Rightarrow \varepsilon^* \le \frac{1}{2}$$

- The maximum optimal error in a two-class classification problem is 0.5.
- Also,

$$\varepsilon^* = \frac{1}{2} \Leftrightarrow E[|2\eta(\boldsymbol{X})-1|] = 0 \Leftrightarrow \eta(\boldsymbol{X}) = \frac{1}{2} \text{ with probability 1.}$$

- A problem, where Bayes error is 0.5 or very close to it, is hopeless.

Bayes Error – Proof (Contd.)

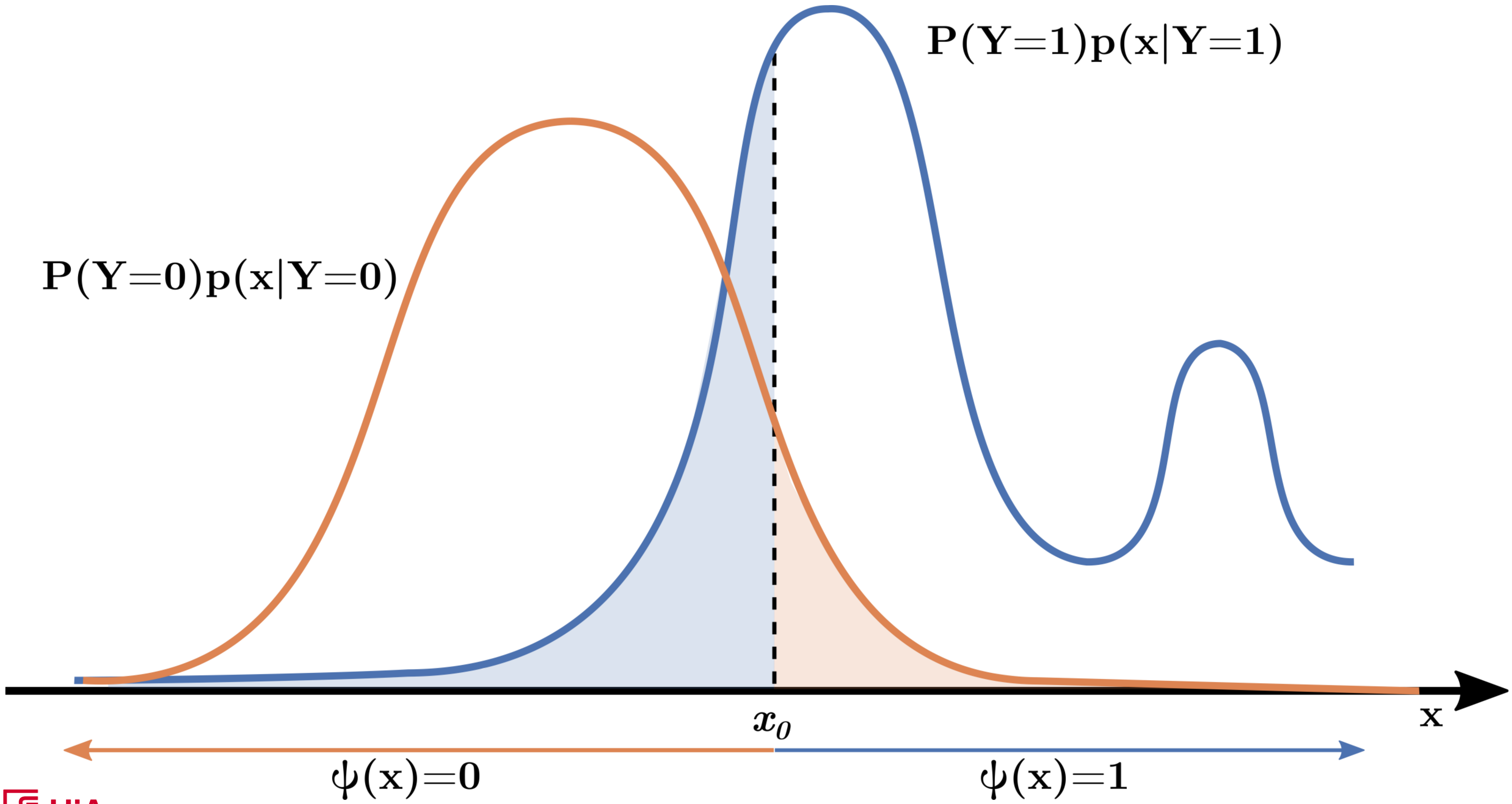
- Recall that for $x \in R^d$:

$$\psi^*(\boldsymbol{x}) = \begin{cases} 1, & P(Y = 1)p(\boldsymbol{x} \mid Y = 1) > P(Y = 0)p(\boldsymbol{x} \mid Y = 0) \\ 0, & \text{otherwise} \end{cases}$$

- Applying (1) above:

$$\begin{aligned} \varepsilon[\psi^*] &= P(Y = 0)\varepsilon^0[\psi^*] + P(Y = 1)\varepsilon^1[\psi^*] \\ &= \int_{\{P(Y=1)p(\boldsymbol{x}|Y=1) > P(Y=0)p(\boldsymbol{x}|Y=0)\}} P(Y = 0)p(\boldsymbol{x}|Y = 0)d\boldsymbol{x} \\ &\quad + \int_{\{P(Y=1)p(\boldsymbol{x}|Y=1) \le P(Y=0)p(\boldsymbol{x}|Y=0)\}} P(Y = 1)p(\boldsymbol{x}|Y = 1)d\boldsymbol{x} \end{aligned}$$

- Bayes error is the sum of integrals of prior-weighted densities over the regions of overlap between them.


P(Y=1)p(x|Y=1)
P(Y=0)p(x|Y=0)
x
x_0
ψ(x)=0
ψ(x)=1

Bayes Error – Transformed feature vector

- Feature transformation: preprocessing, normalization, dimensionality reduction, etc.
- **Theorem**: Given a transformation between feature spaces $t: \mathbb{R}^p \to \mathbb{R}^d$ such that it transforms the original feature vector $\boldsymbol{X} \in \mathbb{R}^p$ as $\boldsymbol{X}' = t(\boldsymbol{X}) \in \mathbb{R}^d$ and the Bayes errors $\varepsilon^*(\boldsymbol{X}, Y)$ and $\varepsilon^*(\boldsymbol{X}', Y)$:
$$\varepsilon^*(\boldsymbol{X}', Y) \geq \varepsilon^*(\boldsymbol{X}, Y)$$
with equality if *t* is reversible.

Minimax classification

- Bayes classifier:

$$\psi^*(\boldsymbol{x}) = \begin{cases} 1, & D^*(\boldsymbol{x}) > k^* \\ 0, & \text{otherwise} \end{cases}$$

  where the **optimal discriminant** $D^*: \mathbb{R}^d \to \mathbb{R}$ is given by

$$D^*(\boldsymbol{x}) = \ln \frac{p(\boldsymbol{x} \mid Y = 1)}{p(\boldsymbol{x} \mid Y = 0)}$$

  for $x \in R^d$, with **optimal threshold** k^* given by

$$k^* = \ln \frac{P(Y = 0)}{P(Y = 1)}$$

- In some cases, prior probabilities are unknown or cannot be estimated reliably.
- For a classifier ψ, the class specific error rates, $\varepsilon^0[\psi]$ and $\varepsilon^1[\psi]$ are independent of prior probabilities. However, overall error rate $\varepsilon[\psi]$ is not available.
- Also available is optimal threshold D^*.
- Optimal threshold k^* however, is not available.

Minimax classification

- Classifiers have to be defined as a function of k :

$$\psi_k^*(\boldsymbol{x}) = \begin{cases} 1, & D^*(\boldsymbol{x}) > k \\ 0, & \text{otherwise} \end{cases} \text{ for } k \in R$$

- Class specific error rates $\varepsilon^0(k) = \varepsilon^0[\psi_k^*]$ and $\varepsilon^1(k) = \varepsilon^1[\psi_k^*]$.
- Each classifier ψ_k^* is a Bayes classifier for a particular set of prior probabilities.
- The Bayes error is a linear combination of $\varepsilon^0(k)$ and $\varepsilon^1(k)$
- Increasing k decreases $\varepsilon^0(k)$ and increases $\varepsilon^1(k)$ and vice versa:
  - A higher k raises the threshold for classifying data as class 1, reducing false positives (fewer class 0 instances misclassified as 1).
  - A higher k makes it harder to classify data as class 1, increasing false negatives (more class 1 instances misclassified as 0).
- A better discriminant should have uniformly smaller $\varepsilon^0(k)$ and $\varepsilon^1(k)$.

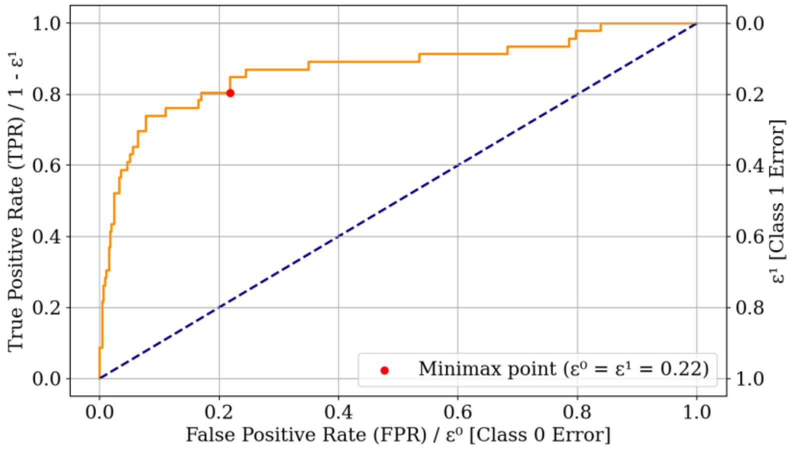
Minimax classification

- A plot of $1 - \varepsilon^1(k)$ vs. $\varepsilon^0(k)$ is called **receiver operating characteristics (ROC)** can be used to assess discriminatory power.
- The area under the curve called **area under ROC curve (AUC)** can help evaluate the performance of the discriminant.
- Since ε^* is a linear combination of $\varepsilon^0(k)$ and $\varepsilon^1(k)$, the maximum value the Bayes error can take is $\max\{\varepsilon^0(k), \varepsilon^1(k)\}$. The **minimax criterion** k_{mm}, such that:

$$k_{mm} = \arg\min_k [\max \varepsilon^0(k), \varepsilon^1(k)\ ]$$

- Thus the **minimax classifier**, becomes:

$$\psi^*_{k_{mm}}(\boldsymbol{x}) = \begin{cases} 1, & D^*(\boldsymbol{x}) > k_{mm} \\ 0, & \text{otherwise} \end{cases}$$

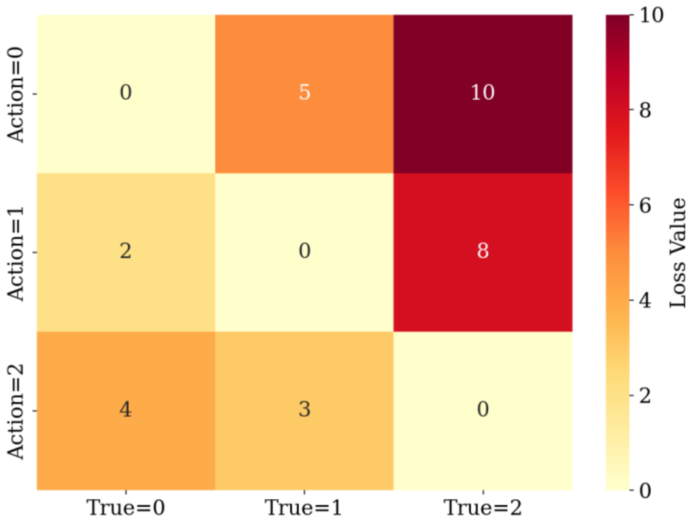


Bayes Decision Theory

- A general procedure for decision making, where optimal classification is a special case.
- Suppose we are building a classifier for an online medical diagnosis site like WebMD.
- There are three **states of nature** (classes, $c = 3$) a user can be in $Y \in \{y_0, y_1, y_2\}$:
  - 0: Common cold
  - 1: Pneumonia
  - 2: Lung Cancer
- The system can take one of three **actions** $\alpha(\boldsymbol{x}) \in \{\alpha_0, \alpha_1, \alpha_2\}(a = 3)$:
  - Diagnosing the user to have diseases 0, 1, or 2 based on $\alpha_0, \alpha_1, \alpha_2$
  - There could be a case where, $a > c$.
- The user can upload their body temperature $\boldsymbol{x}_1$, a pain index $\boldsymbol{x}_2$.
- These features are used to compute posterior probabilities $P(Y = j \mid X = x)$

Bayes Decision Theory

- Now we need to evaluate how well the system works: a **loss function** λ_{ij} which is the cost of taking action α_i (diagnosing disease y_i) when actually the user had a disease y_j.
- We decide on a **loss matrix** as shown in the figure:
    - Diagnosing lung cancer as common cold incurs high penalty.
    - Diagnosing common cold as lung cancer incurs moderate penalty.
- Suppose we have some data from the past based on the reported body temperature what kind of disease patients have had.



Bayes Decision Theory

- We calculate the posterior probabilities using **Bayes Theorem**:

$$P(Y = y_i \mid \boldsymbol{X} = \boldsymbol{x}) = \frac{P(\boldsymbol{X} = \boldsymbol{x} \mid Y = y_i)P(Y = Y_i)}{P(\boldsymbol{X} = \boldsymbol{x})}$$

- Suppose we know these posterior probabilities for a feature $\boldsymbol{x}_1$:

$P(Y = 0 \mid \boldsymbol{x}_1) = 0.2$, $P(Y = 1 \mid \boldsymbol{x}_1) = 0.3$, and $P(Y = 2 \mid \boldsymbol{x}_1) = 0.5$

- First, we need to know what would be the risk involved with taking each action, called **conditional risk** $\alpha(\boldsymbol{x})$:
  - α_0: $R(\alpha_0) = (0 \cdot 0.2) + (5 \cdot 0.3) + (10 \cdot 0.5) = 6.5$
  - α_1: $R(\alpha_1) = (2 \cdot 0.2) + (0 \cdot 0.3) + (8 \cdot 0.5) = 4.4$
  - α_2: $R(\alpha_2) = (4 \cdot 0.2) + (3 \cdot 0.3) + (0 \cdot 0.5) = 1.7$
- Bayes decision rule is that it is enough to select an action with minimum conditional risk, hence $\alpha(\boldsymbol{x}) = \alpha_2$.

Bayes Decision Theory

Generalizing:

- On observing the feature vector $\boldsymbol{X} = \boldsymbol{x}$, and taking an **action** $\alpha(\boldsymbol{x}) \in \{\alpha_0, \alpha_1, \dots, \alpha_{a-1}\}$
- Assuming there are c **states of nature** such that $Y \in \{y_0, y_1, \dots, y_{c-1}\}$.
- And each action incurs a **loss** $\lambda_{ij} = \text{cost of taking action } \alpha_i \text{ when true state of nature is } j$.
- Where the action α_i might be simply deciding the true state of nature is i $(a = c)$ or $a > c$, where there maybe an extra action of rejecting to take an action.
- The conditional loss upon observing a feature vector $\boldsymbol{X} = \boldsymbol{x}$, called **conditional risk** is given by:

$$R[\alpha(\boldsymbol{x}) = \alpha_i] = \sum_{j=0}^{c-1} \lambda_{ij} P(Y = j \mid \boldsymbol{X} = \boldsymbol{x})$$

- The **risk** over the entire feature space $\boldsymbol{x} \in R^d$ is given by:

$$R = E\left[R\big(\alpha(\boldsymbol{X})\big)\right] = \int_{\boldsymbol{x} \in R^d} R\big(\alpha(\boldsymbol{x})\big) p(\boldsymbol{x})\, d\boldsymbol{x}$$

- To minimize R, it is enough to select an action $\alpha(\boldsymbol{x}) = \alpha_i$ such that the conditional risk $R[\alpha(\boldsymbol{x}) = \alpha_i]$ is minimum at each value $\boldsymbol{x} \in R^d$. This strategy is called the **Bayes decision rule**, with corresponding optimal **Bayes risk** R^*.

Bayes Decision Theory – Special Case

- Now, if $a = c = 2$:

$$R[\alpha(\boldsymbol{x}) = \alpha_0] = \lambda_{00}P(Y = 0 \mid \boldsymbol{X} = \boldsymbol{x}) + \lambda_{01}P(Y = 1 \mid \boldsymbol{X} = \boldsymbol{x})$$
$$R[\alpha(\boldsymbol{x}) = \alpha_1] = \lambda_{10}P(Y = 0 \mid \boldsymbol{X} = \boldsymbol{x}) + \lambda_{11}P(Y = 1 \mid \boldsymbol{X} = \boldsymbol{x})$$

- If $R[\alpha(\boldsymbol{x}) = \alpha_0] < R[\alpha(\boldsymbol{x}) = \alpha_1]$, we decide action α_0 if:

$$R[\alpha(\boldsymbol{x}) = \alpha_1] > R[\alpha(\boldsymbol{x}) = \alpha_0]$$

$$\lambda_{10}P(Y = 0 \mid \boldsymbol{X} = \boldsymbol{x}) + \lambda_{11}P(Y = 1 \mid \boldsymbol{X} = \boldsymbol{x}) > \lambda_{00}P(Y = 0 \mid \boldsymbol{X} = \boldsymbol{x}) + \lambda_{01}P(Y = 1 \mid \boldsymbol{X} = \boldsymbol{x})$$

$$(\lambda_{10} - \lambda_{00})\, P(Y = 0 \mid \boldsymbol{X} = \boldsymbol{x}) > (\lambda_{01} - \lambda_{11})P(Y = 1 \mid \boldsymbol{X} = \boldsymbol{x})$$

Applying Bayes Theorem and assuming $\lambda_{10} > \lambda_{00}$ and $\lambda_{01} > \lambda_{11}$:

$$\frac{p(\boldsymbol{x} \mid Y = 0)}{p(\boldsymbol{x} \mid Y = 1)} > \frac{\lambda_{01} - \lambda_{11}}{\lambda_{10} - \lambda_{00}} \frac{P(Y = 1)}{P(Y = 0)}$$

Bayes Decision Theory – Special Case

- The loss $\lambda_{00} = \lambda_{11} = 0$ and $\lambda_{10} = \lambda_{01} = 1$ is called the **0-1 loss**, equivalent to the misclassification loss.
- With these losses, $\frac{p(\boldsymbol{x}|Y=0)}{p(\boldsymbol{x}|Y=1)} > \frac{\lambda_{01}-\lambda_{11}}{\lambda_{10}-\lambda_{00}} \frac{P(Y=1)}{P(Y=0)}$ becomes $\frac{p(\boldsymbol{x}|Y=0)}{p(\boldsymbol{x}|Y=1)} > \frac{P(Y=1)}{P(Y=0)}$, which is equivalent to the Bayes classifier.
- Thus, if the action is simply deciding the state of nature is $i \; \forall \; i \in \{0,1\}$, the 0-1 loss case leads to the **optimal binary classification**.
- The (conditional) risk reduces to (conditional) classification error.
- The Bayes decision rule reduces to Bayes classifier.
- The Bayes risk reduces to Bayes error.

Thank You